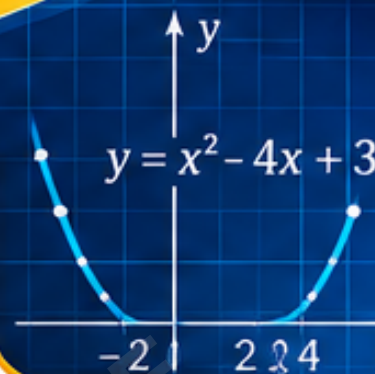


Grade 11 Core MATHEMATICS BOOK 1

Concepts, Notes & Application



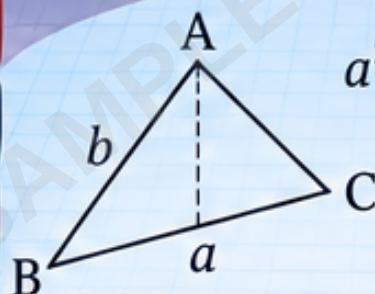
Algebra

Geometry

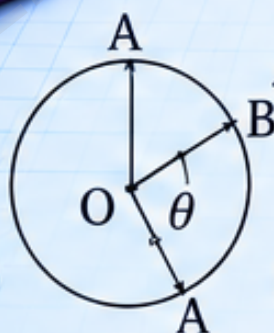
Trigonometry

Calculus

Statistics



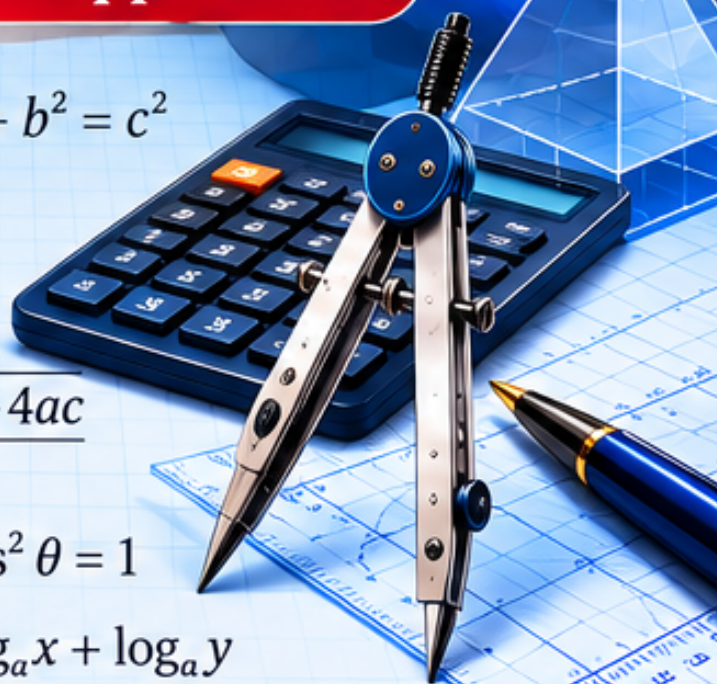
$$a^2 + b^2 = c^2$$



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\log_a(xy) = \log_a x + \log_a y$$



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GRADE 11 CORE MATHEMATICS REVISION BOOK

BOOK 1 - CONCEPTS, NOTES & APPLICATION

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Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose guidance, sacrifices and belief in education continue to inspire my journey and the work I do as an educator.

To my mother, Hyrine Kerubo, for her love, strength, prayers and unwavering support.

To my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe, for their guidance, encouragement and the important role they have played in my journey.

To my friends Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob and Tim, and to Mr Peter Kimani, for the encouragement, friendship and support shared along the way.

To Her, my special one, for the encouragement, patience and inspiration.

To my fellow teachers, whose commitment to education continues to inspire me, and to every learner who keeps trying even when Mathematics feels difficult: may every question you attempt take you one step closer to becoming the achiever you are capable of being.

To each of you, this book carries a part of your contribution to my journey.

- Hezekiah Omweri Nyatuka

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How to Use This Book

The Peak Achievers learning path is: Understand -> Practise -> Apply -> Revise -> Master. Work through each chapter, attempt the exercises before checking answers, complete the strand revisions, then progress to Mixed Concepts and the six final practice tests.

- HINT - a short clue to help you begin.
- REMEMBER - a key rule or formula.
- CAUTION - a common error to avoid.
- THINK BIG - reasoning beyond routine calculation.
- SPOTLIGHT - an important examination or application idea.

Book Structure

- Strand 1: Numbers & Algebra - Chapters 1.1-1.7
- Strand 2: Measurements & Geometry - Chapters 2.1-2.6
- Strand 3: Statistics & Probability - Chapters 3.1-3.2

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- Strand 4: Calculus - Chapter 4.1
- Four Mixed Concepts sets
- Six 100-mark final practice tests with marking schemes

FREE SAMPLE - NOT FOR SALE

Chapter 1.1 - Quadratic Expressions and Equations

Learning Outcomes

- Understand and apply the key ideas in Quadratic Expressions and Equations.
- Use correct mathematical notation, methods and reasoning.
- Solve routine and real-life application problems.
- Check answers for reasonableness and communicate working clearly.

Core Ideas

A quadratic expression has highest power 2. A quadratic equation has the form $ax^2 + bx + c = 0$, where $a \neq 0$.

$$ax^2 + bx + c = 0$$

REMEMBER: Always write the equation in standard form before choosing a method.

Factorisation and Completing the Square

Example: $x^2 - 7x + 12 = 0$ gives $(x - 3)(x - 4) = 0$, hence $x = 3$ or 4 .

Example: $x^2 + 6x - 7 = 0$ gives $(x + 3)^2 = 16$, hence $x = 1$ or -7 .

Quadratic Formula and Discriminant

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$D = b^2 - 4ac$$

If $D > 0$ there are two distinct real roots; if $D = 0$ there is one repeated real root; if $D < 0$ there are no real roots.

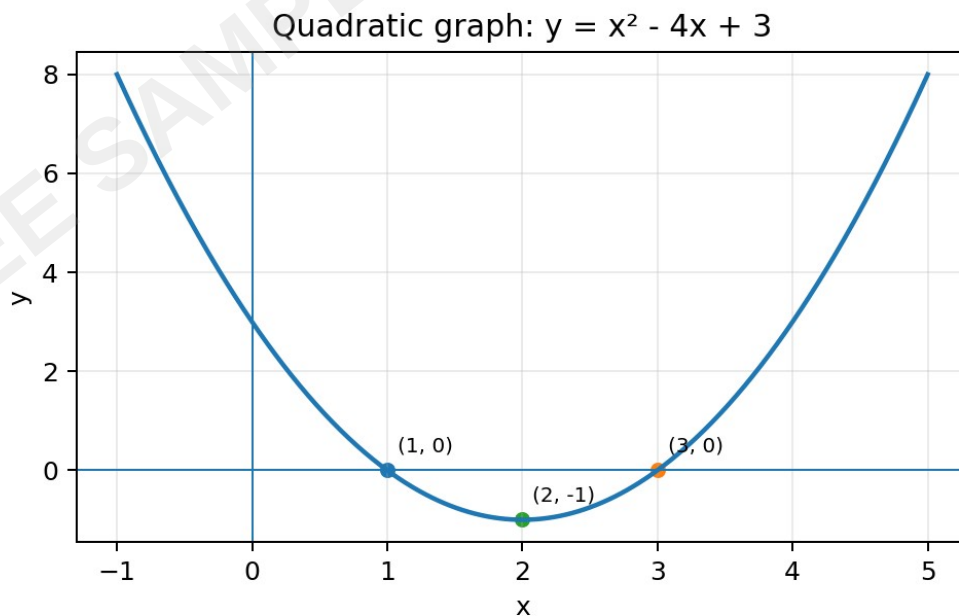


Figure: a quadratic graph and its roots/turning point.

Application

A rectangular garden has area 96 m^2 and its length is 4 m greater than its width. Let width be x : $x(x+4)=96$. Hence $x^2+4x-96=0$, giving width 8 m and length 12 m.

THINK BIG: Quadratic equations often arise when an unknown quantity is multiplied by another expression containing the same unknown.

Topical Exercise and Answers

1. Solve $x^2 - 11x + 28 = 0$.

Solution:

Step 1: Identify the mathematical concept required by the question.

Step 2: State the relevant definition, rule or formula.

Step 3: Apply it directly to the information given and simplify where necessary.

Final answer: $x = 4$ or 7 .

Answer: $x = 4$ or 7 .

2. Solve $2x^2 + x - 6 = 0$.

Solution:

Step 1: Identify the mathematical concept required by the question.

Step 2: State the relevant definition, rule or formula.

Step 3: Apply it directly to the information given and simplify where necessary.

Final answer: $x = 3/2$ or -2 .

Answer: $x = 3/2$ or -2 .

3. Solve $x^2 + 8x + 7 = 0$.

Solution:

Step 1: Identify the mathematical concept required by the question.

Step 2: State the relevant definition, rule or formula.

Step 3: Apply it directly to the information given and simplify where necessary.

Final answer: $x = -1$ or -7 .

Answer: $x = -1$ or -7 .

4. Solve $3x^2 - 7x + 2 = 0$.

Solution:

Step 1: Identify the mathematical concept required by the question.

Step 2: State the relevant definition, rule or formula.

Step 3: Apply it directly to the information given and simplify where necessary.

Final answer: $x = 1/3$ or 2 .

Answer: $x = 1/3$ or 2 .

5. State the nature of roots of $2x^2+4x+7=0$.

Solution:

Step 1: Identify a , b and c from $ax^2 + bx + c = 0$.

Step 2: Use $D = b^2 - 4ac$.

Step 3: Substitute the coefficients from the question and simplify the arithmetic.

Step 4: Interpret the sign of D : $D > 0$ two real roots, $D = 0$ repeated root, $D < 0$ no real roots.

Final answer: $D = -40$, so no real roots.

Answer: $D = -40$, so no real roots.

6. Two consecutive positive integers have product 156. Find them.

Solution:

Step 1: Identify the mathematical concept required by the question.

Step 2: State the relevant definition, rule or formula.

Step 3: Apply it directly to the information given and simplify where necessary.

Final answer: 12 and 13.

Answer: 12 and 13.

7. Rectangle area 180 m^2 ; length is 3 m more than width.

Solution:

Step 1: Identify the mathematical concept required by the question.

Step 2: State the relevant definition, rule or formula.

Step 3: Apply it directly to the information given and simplify where necessary.

Final answer: 12 m by 15 m.

Answer: 12 m by 15 m.

Chapter 1.2 - Irrational Numbers and Surds

Learning Outcomes

- Understand and apply the key ideas in Irrational Numbers and Surds.
- Use correct mathematical notation, methods and reasoning.
- Solve routine and real-life application problems.
- Check answers for reasonableness and communicate working clearly.

Rational and Irrational Numbers

A rational number can be written p/q where p and q are integers and $q \neq 0$. Irrational numbers cannot be written exactly in that form. Surds are irrational roots left in exact form.

Simplifying and Operating with Surds

$$\sqrt{72} = 6\sqrt{2}$$

Like surds can be added or subtracted: $\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$.

Multiplication: $(3\sqrt{2})(2\sqrt{6}) = 12\sqrt{3}$.

Conjugates and Rationalisation

The conjugate of $a+b\sqrt{c}$ is $a-b\sqrt{c}$. Example: $4/(3+\sqrt{5}) = 3-\sqrt{5}$.

CAUTION: Do not combine unlike surds by adding the numbers inside the radical.

Geometry Application

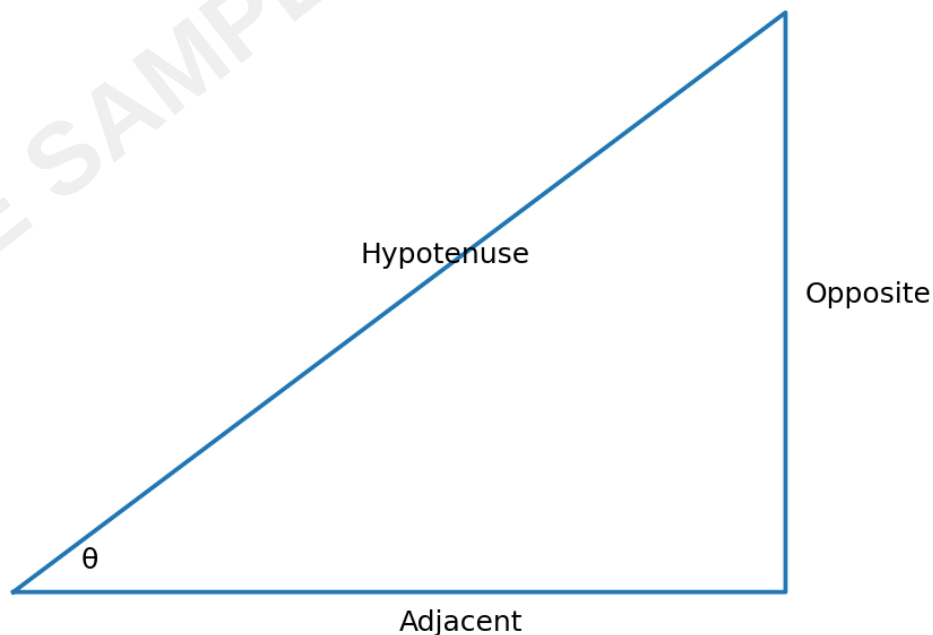


Figure: right-triangle geometry can produce irrational lengths.

For perpendicular sides 5 and 7, the hypotenuse is $\sqrt{74}$.

Step 3: Lower bound = stated value - half-unit; upper bound = stated value + half-unit.

Step 4: For relative/percentage error, divide absolute error by the measured value and multiply by 100% when required.

Final answer: $2650 \leq d < 2750$.

Answer: $2650 \leq d < 2750$.

8. Rectangle 12.4×7.2 nearest 0.1 lower area.

Solution:

Step 1: Identify the rounding unit used in the stated measurement.

Step 2: Half of that unit gives the maximum rounding error.

Step 3: Lower bound = stated value - half-unit; upper bound = stated value + half-unit.

Step 4: For relative/percentage error, divide absolute error by the measured value and multiply by 100% when required.

Final answer: 88.3025.

Answer: 88.3025.

9. Limiting upper area.

Solution:

Step 1: Identify the rounding unit used in the stated measurement.

Step 2: Half of that unit gives the maximum rounding error.

Step 3: Lower bound = stated value - half-unit; upper bound = stated value + half-unit.

Step 4: For relative/percentage error, divide absolute error by the measured value and multiply by 100% when required.

Final answer: 90.2625.

Answer: 90.2625.

10. 250km nearest 1, 4.0h nearest 0.1: speed bounds.

Solution:

Step 1: Identify the rounding unit used in the stated measurement.

Step 2: Half of that unit gives the maximum rounding error.

Step 3: Lower bound = stated value - half-unit; upper bound = stated value + half-unit.

Step 4: For relative/percentage error, divide absolute error by the measured value and multiply by 100% when required.

Final answer: $\approx 61.60 \leq v < 63.42$ km/h.

Answer: $\approx 61.60 \leq v < 63.42$ km/h.